

Craniotomy

A craniotomy is an operation where the surgeon removes part or all of the brain tumour.

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What is a craniotomy?

To remove part or all of the [brain tumour](#), you usually have an operation called a craniotomy.

You usually have the surgery under a general anaesthetic. The surgeon removes a piece of skull over the tumour to make an opening. They use a powerful microscope and sometimes scans to look at the brain. This means they can carefully remove the tumour without taking away healthy areas of [the brain](#). After they have removed part or all of the tumour, the surgeon replaces the piece of skull. They will usually fix it place with small titanium plates.

Sometimes the surgeon can remove the tumour through a very small opening in the skull using a neuroendoscope. This is a thin, flexible tube with a camera on 1 end and an eyepiece on the other. This type of surgery is sometimes called endoscopic or keyhole surgery.

For some types of tumours affecting the [pituitary gland](#) or at the base of the skull, this surgery can also be done by passing the endoscope up the nose into the brain.

Your surgeon will explain which type of surgery is best for you.

If it is not possible to remove the whole tumour, the surgeon will remove as much as they can. This is called a partial or sub-total resection, or debulking.

After the operation, the tissue is sent to a laboratory for tests.

5-ALA (Gliolan)

Some people have a drug called 5-ALA (Gliolan) as a drink before the operation. It is also called 'the pink drink' or photodynamic diagnosis (PDD). During surgery, 5-ALA makes brain tumour cells glow pink or red under a blue light. This helps the surgeon decide which areas to remove. It is not used in all brain tumour operations. This is because some types of tumours will not glow with this drink. If your surgeon feels it would be helpful, they will discuss it with you.

5-ALA makes your eyes and skin sensitive to light for up to 24 hours after taking it. During this time, it is important to avoid direct sunlight and brightly focused indoor light. While you are in hospital, the staff will make sure you are protected from bright, direct light.

Craniotomy while you are awake (awake craniotomy)

Sometimes the surgeon will suggest a craniotomy while you are awake. This may be offered if they think that keeping you awake during the operation may make it possible to:

- remove more of the tumour
- remove it more safely.

This is instead of general anaesthetic, when you are put to sleep for the whole operation.

This may sound frightening, but people usually cope with it well. You will have lots of chances to discuss the risks and benefits and any worries you may have with the neurosurgical team. This includes doctors, nurses and psychologists.

Because you are awake, the surgical team can ask you questions and check that your speech and movement are not being affected by the surgery. This reduces the risk of damage to your brain. It also means the surgeon may be able remove more of the tumour. If there are changes to your speech or thinking, the surgeon stops operating and assesses the situation.

This surgery can be done with sedation or a general anaesthetic for the first part of the surgery. This is when the surgeon makes an opening through the skull. The surgeon uses local anaesthetic injections to numb the scalp. They gently wake you when they are ready to operate on the brain. You should not feel any pain during the operation. This is because the brain has no nerve endings.

Some surgeons may suggest doing the operation with you awake throughout. They will numb the scalp and head with local anaesthetic injections.

You can see and talk to the doctors and nurses during the operation. You can tell them straight away if you are worried about anything that is happening.

How diagnosis affects your right to drive

Following diagnosis and treatment for a brain tumour, most people will not be allowed to drive for a period of time. If you drive, it is important to discuss with your doctor how your diagnosis and treatment for a brain tumour affects your right to drive.

If you have a driving licence, you must tell the licencing agency ([DVLA](#) or [DVA](#)) that you have been diagnosed with a brain tumour. We have more information about how a [brain tumour may affect your right to drive](#).

About our information

This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer.

References



Below is a sample of the sources used in our primary brain tumour information. If you would like more information about the sources we use, please contact us at informationproductionteam@macmillan.org.uk

EANO-ESMO Clinical Practice Guidelines for prophylaxis, diagnosis, treatment and follow-up: Neurological and vascular complications of primary and secondary brain tumours. 2021. Available from www.eano.eu/publications/eano-guidelines/eano-esmo-clinical-practice-guidelines-for-prophylaxis-diagnosis-treatment-and-follow-up-neurological-and-vascular-complications-of-primary-and-secondary-brain-tumours [accessed August 2024].

NICE Guideline NG99. Brain tumours (primary) and brain metastases in over 16s. 2018 (updated 2021). Available from: www.nice.org.uk/guidance/ng99 [accessed August 2024].

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Our cancer information meets the [PIF TICK](#) quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about [how we produce our information](#).

The language we use

We want everyone affected by cancer to feel our information is written for them.

We want our information to be as clear as possible. To do this, we try to:

- use plain English
- explain medical words
-

use short sentences

- use illustrations to explain text
- structure the information clearly
- make sure important points are clear.

We use gender-inclusive language and talk to our readers as 'you' so that everyone feels included. Where clinically necessary we use the terms 'men' and 'women' or 'male' and 'female'. For example, we do so when talking about parts of the body or mentioning statistics or research about who is affected.

You can read more about how we produce our information [here](#).

How we can help

Macmillan Support Line

The Macmillan Support Line is a free and confidential service for people living with and affected by cancer. If you need to talk, we'll listen.



Call us free on 0808 808 0000

8am to 8pm, 7 days a week. You can talk to our cancer specialists about anything.



Email the Macmillan Support Line

Email the Macmillan Support Team if you have any questions.

Chat online

Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don't want to call.



Online Community

Chat online anonymously to others who understand what you are going through. Our community is available 24/7 and has dedicated forums where you can get advice and ask our experts.



Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.

Get help